Felipe Arancibia Carrasco

AI Engineer

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PROFILE

Electronic civil engineer with a Master of Science in Electronic Engineering from Universidad Técnica Federico Santa María. Since 2021 I have built computer vision and AI solutions for industrial and security settings, from applied research to deployment under latency constraints. My current research focus is LLM-based agentic systems.

WORK EXPERIENCE

AI Engineer · VSAAS — Jan 2026 – Present
Viña del Mar, Chile

- R&D on new automatic-analysis capabilities for intelligent video-surveillance platforms, using real VSAAS infrastructure as the validation environment.
- Design and experimental evaluation of agentic architectures that combine perception (computer vision), contextual memory and LLM-based reasoning for autonomous event management.
- Training, fine-tuning and comparative evaluation of computer vision and deep learning models in security and critical-infrastructure scenarios.
- Python inference services and APIs (FastAPI/gRPC) to integrate AI solutions into production environments with latency constraints.
- Work with intelligent agents: context management, tool calling and response automation.

AI Engineer · ETT Transferencia de Tecnologías SpA — Jan 2025 – Nov 2025
Viña del Mar, Chile

- Applied research and development of computer vision models to optimize grinding and crushing processes in dynamic, noisy industrial environments.
- Automated counting of grinding balls and continuous monitoring of their geometry and wear, using object detection and multi-object tracking; load optimization in SAG mills.
- Estimation of the internal load of SAG mills using computer vision and 3D reconstruction from image sequences.
- Extraction, processing and analysis of data from industrial measurement systems.
- Integration and deployment of AI solutions in real operating environments.

AI Engineer · Kauel — Jun 2021 – Jul 2024
Santiago, Chile

- Development of distributed (Edge/Cloud) computer vision solutions for industrial applications with latency constraints.
- Design of automated volume-estimation services using image preprocessing pipelines and deep networks; error margin ≤ 5% in production.
- Training, validation and deployment of machine learning models, including real-time inference.
- Development and integration of APIs and Docker containerization for deploying models to production.
- Data processing and analysis; technical documentation and implementation support.

RESEARCH AND PUBLICATIONS

RESEARCH LINE · IN PROGRESS

AI-based agentic systems: memory, reasoning and autonomous event management

Study of agentic architectures that combine visual perception (vision-language models), contextual memory and LLM-based reasoning for security and critical-infrastructure systems. State-of-the-art review: ReAct, CoAgent, ShowUI and Model Context Protocol (MCP).

PROPOSAL

AI-based agentic systems for intelligent event management in video surveillance

Architecture integrating perception (computer vision), dual short- and long-term memory, LLM reasoning and planning; validation on the VSAAS platform with real security cases.

PUBLICATION · IEEE CHILECON, 2019

Parameter Estimation over Wired Networks with Time-Delay and Packet Collision

Arancibia, F., Carvajal, R. and Agüero, J. C. IEEE Chilean Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON), Valparaíso. DOI: 10.1109/CHILECON47746.2019.8987445.

EDUCATION

M.Sc. in Electronic Engineering · Universidad Técnica Federico Santa María 2018 – 2020

Valparaíso, Chile

Machine learning and artificial intelligence · Stochastic processes and statistical modeling · Estimation theory and optimization algorithms · Signal processing and pattern recognition · Advanced automatic control · Modeling and simulation of dynamic systems. EQF level 7.

Electronic Civil Engineer · Universidad Técnica Federico Santa María 2013 – 2020

Valparaíso, Chile

Digital signal processing and instrumentation · Automatic control and industrial automation · Software development · Analog and digital circuits · Embedded systems and microcontrollers · Telecommunications and data transmission. EQF level 7.

SKILLS

Agentic AI and LLMs (reasoning, planning, tool calling) · Dual spatio-temporal memory · Computer vision and multi-object tracking · Applied research and experimental design · Python, PyTorch, Docker, FastAPI/gRPC · Edge computing and real-time deployment · Teamwork